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Botany and Anatomical Structures

Course Code: 2121

Semester: 2

Credits: 3

Wood and Paper Technology

A comprehensive guide to plant diversity, taxonomy, and the internal anatomy of wood-producing plants.

Course Overview

Category	Engineering Science
Teaching Scheme	3 Hours Lecture per week (L:3, T:0, P:0)
Total Periods	45 per semester
Assessment	CIE (50 Marks) + ESE (50 Marks)

Course Objectives

- Understand plant kingdom classifications.
- Identify plants with economic and industrial importance.

- Analyze the internal structure and reproduction mechanisms of plants.

Course Outcomes (COs)

CO No.	Description	Hours	Cognitive Level
CO1	Summarize plant diversity	8	Understanding
CO2	Describe plant taxonomy and morphology	13	Understanding
CO3	Illustrate plant anatomy and tissues	12	Understanding
CO4	Discuss plant body structure and wood anatomy	10	Understanding

8 Hours

Bryophytes	"Amphibians of plant kingdom". Lack true roots/vascular tissues.	Peat formation (Sphagnum), soil erosion prevention.
Pteridophytes	First terrestrial plants to possess vascular tissues (Xylem/Phloem).	Medicinal use, ornamental plants.
Gymnosperms	Plants with naked seeds (ovules not enclosed by ovary wall).	Timber (Pine, Cedar), resin, turpentine.

1.3 Genetics and Diversity

Diversity arises from changes in the genetic material (DNA).

- **Nucleus:** The control center containing genetic material.
- **DNA & Genes:** DNA is the hereditary material. Genes are segments of DNA that code for specific proteins.
- **Gene Action:** The process by which a gene's information is used to synthesize a functional gene product (protein).

Module 2: Plant Taxonomy and Morphology

13 Hours

2.1 Morphology of Flowering Plants

Morphology deals with the external form and structure of plants.

- **Root Systems:**
 - Tap root (Dicot)
 - Fibrous root (Monocot)
 - Adventitious root (Roots arising from parts other than radicle)
- **Leaves:** Parts include Lamina, Petiole, and Stipules.
 - *Simple leaf*: Single undivided blade.
 - *Compound leaf*: Blade divided into leaflets.
 - *Phyllotaxy*: Pattern of arrangement of leaves on stem (Alternate, Opposite, Whorled).

2.2 Inflorescence and Flower

Inflorescence: The arrangement of flowers on the floral axis.

- **Racemose:** Main axis continues to grow; flowers in acropetal succession.
- **Cymose:** Main axis terminates in a flower; flowers in basipetal succession.

Flower Parts: Calyx (Sepals), Corolla (Petals), Androecium (Stamens), Gynoecium (Carpels).

2.3 Angiosperm Taxonomy

Binomial Nomenclature: Scientific naming using two words (Genus and Species). Example: *Tectona grandis* (Teak).

Key Families for Wood Technology:

1. **Verbenaceae:** Example: *Tectona grandis* (Teak) - Essential for high-quality timber.
2. **Fabaceae:** Example: *Acacia* - Used for furniture and tannin.
3. **Myrtaceae:** Example: *Eucalyptus* - Primary source for paper pulp.
4. **Euphorbiaceae:** Example: *Hevea brasiliensis* (Rubber).

Module 3: Plant Anatomy and Tissues

12 Hours

3.1 Plant Cell Ultrastructure

The cell wall is the most critical part for wood technology. It consists of cellulose, hemicellulose, and lignin.

- **Pits:** Thin areas in the cell wall where secondary wall is not deposited, allowing transport.
- **Plasmodesmata:** Cytoplasmic channels between adjacent cells.

3.2 Plant Tissues

Meristematic Tissues: Actively dividing cells. Found at Root apex and Shoot apex.

Permanent Tissues:

- *Simple:* Parenchyma, Collenchyma, Sclerenchyma.
- *Complex:* Xylem (water transport) and Phloem (food transport).

3.3 Vascular Bundles

Arrangement of Xylem and Phloem.

- **Conjoint:** Xylem and Phloem on the same radius.
- **Collateral:** Phloem outside, Xylem inside.
- **Open:** Cambium present (allows secondary growth).
- **Closed:** Cambium absent (no secondary growth).

Module 4: Wood Anatomy

10 Hours

4.1 Primary and Secondary Structure

Primary growth increases length, while secondary growth increases girth (thickness).

Secondary Growth in Dicot Stem: 1. Formation of Cambial Ring. 2. Activity of Cambium producing Secondary Xylem (Wood) and Secondary Phloem. 3. Formation of Periderm (Cork).

4.2 Wood Characteristics

Feature	Description
Growth Rings	Concentric layers of wood produced in one year (Spring wood + Autumn wood).
Heartwood	Central, dark, non-functional, durable part of wood (contains resins/tannins).
Sapwood	Outer, light-colored, functional part for water transport.
Tyloses	Balloon-like outgrowths of parenchyma into xylem vessels; block water flow.

Heartwood (Heartwood) is the non-functional, durable part of wood. It is located in the center of the wood and contains resins/tannins. Sapwood (Sapwood) is the functional part for water transport. It is located on the outer part of the wood.

Formula Bank & Calculations

1. Determining Tree Age (Dendrochronology):

Age = Number of Annual Rings (Growth Rings)

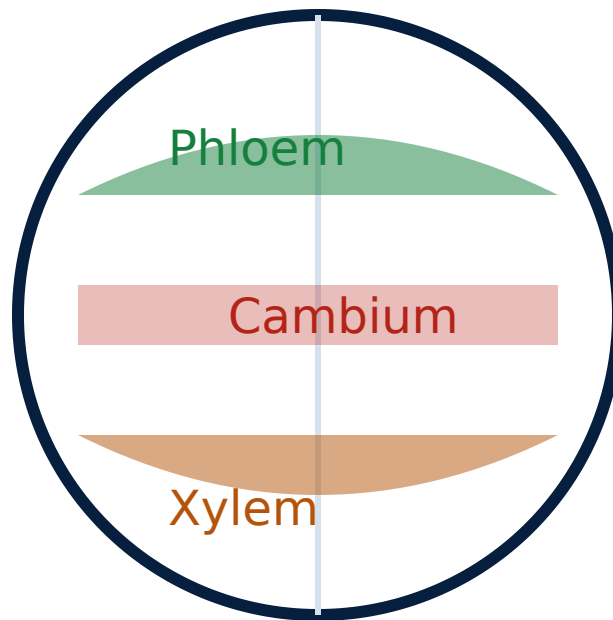
Example: A cross-section of a teak stem shows 45 distinct rings. The estimated age of the tree is 45 years.

Wood Density Calculator

Mass (kg):

Volume (m³):

Visual Library



Typical Conjoint Collateral Open Vascular Bundle (Dicot Stem)

Module-wise Practice Questions

▼ Q: Differentiate between Heartwood and Sapwood. (5 Marks)

Heartwood: Central part, dark color, physiologically inactive, highly durable due to deposition of resins/tannins, resistant to pests.

Sapwood: Peripheral part, light color, physiologically active (transports water), less durable, susceptible to pest attacks.

▼ Q: What are Tyloses and why are they important in wood technology? (3 Marks)

Tyloses are balloon-like extensions of xylem parenchyma cells into the lumen of vessels. They block the vessels, making the wood impermeable. This is important for making wood resistant to decay and for certain industrial uses like barrel making.

▼ Q: Explain the Binomial Nomenclature with an example. (2 Marks)

It is the system of providing a scientific name with two components: Genus and Species. Example: *Mangifera indica* (Mango).

Practice Model Paper

Time: 3 Hours | Max Marks: 50

Part A (Answer all - 5 x 2 = 10 Marks)

1. Define Phyllotaxy.
2. Name the two kingdoms in Linnaeus's classification.
3. What is a Meristem?
4. Define Inflorescence.
5. What is the function of Lenticels?

Part B (Answer any five - 5 x 4 = 20 Marks)

1. Explain the economic importance of Fungi.
2. Describe the parts of a typical flower.
3. Differentiate between Monocot and Dicot stems.
4. Explain the structure of a plant cell wall.
5. Write a note on Herbarium techniques.

Part C (Answer any two - 2 x 10 = 20 Marks)

1. Explain the process of Secondary Growth in Dicot stems with diagrams.
2. Discuss the salient features and economic importance of Gymnosperms.
3. Describe the vegetative and floral characteristics of the family Verbenaceae.

Quick Revision

- **Linnaeus:** 2 Kingdom classification.
- **Eichler:** Phylogenetic classification.
- **Xylem:** Water transport; **Phloem:** Food transport.
- **Cambium:** Responsible for secondary growth (girth).
- **Tectona grandis:** Teak (Verbenaceae).
- **Growth Rings:** Spring wood + Autumn wood.

Glossary

Autotrophic

Organisms that produce their own food.

Dendrochronology

Science of dating based on tree rings.

Lignin

Complex organic polymer that makes wood rigid.

Thallus

Plant body not differentiated into roots, stem, and leaves.

This content is based on the official syllabus for **Course 2121: Botany and Anatomical Structures** (Revision 2021) provided by the **State Institute of Technical Teachers' Training and Research (SITTTR)**, Kerala.

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